Songyue Chen

chensongyue98@ucla.edu | +1 310-437-3465 | Los Angeles, CA | ResearchGate

Ph.D. researcher in Bioengineering at UCLA developing AI-accelerated methods for healthcare through magnetoelastic biosensing, physics-informed inverse modeling, and efficient multimodal generative reconstruction from sparse sensor data.

Research focus: AI Accelerated Science and Engineering Innovation for healthcare; efficient generative AI and inference from sparse biomedical sensor data; physics-informed inverse problems; wearable bioelectronics.

EDUCATION & APPOINTMENTS

University of California, Los Angeles, Los Angeles, CA
Ph.D. Student, Bioengineering; Graduate Student Researcher, Jun Chen Lab

Peking University, Beijing, China
M.S., Pharmaceutical Science

Harbin Medical University, Harbin, China
B.S., Pharmaceutical Science

RESEARCH EXPERIENCE

UCLA | Jun Chen Lab | AI-Accelerated Skin Cancer Mechanomedicine

- Develop soft magnetoelastic microneedle patches integrated with Hall sensor arrays for rapid, noninvasive skin-cancer screening and quantitative tissue-rigidity measurement.
- Build physics-informed inverse models and biomechanical tomography pipelines to infer tumor depth and stiffness from sparse sensor data.
- Develop super-resolution and diffusion-based sensor-to-image generative models for high-resolution stiffness mapping; validate in simulation, phantoms, murine models, and excised human skin against strain elastography and atomic force microscopy.
- Extend related liquid magnetic sensing platforms for ambulatory cardiac monitoring and mechanical / chemical sensing.

Peking University | Wanliang Lu Lab | Microneedle Therapeutics

- Developed a fast-dissolving nanoparticle microneedle patch for obesity therapy; led carrier synthesis, patch fabrication, mechanical characterization, and in vivo evaluation.
- Performed molecular-biology and bioinformatic analyses of therapeutic response, including inflammation and adipocyte-browning readouts.
- Contributed to targeted nanomedicine and virus-like nanoparticle systems for cancer and gene-delivery applications.

Collaborative Project | Generative Protein Modeling

- Co-authored a NeurIPS 2024 workshop paper on generative protein modeling and medical foundation models.

SELECTED PUBLICATIONS & PREPRINTS

- Zhao, X., **Chen, S.**, et al. Soft Magnetoelastic Microneedle Patch for Rapid Skin Cancer Screening. *Nature Sensors* Under review.
- Chen, X., Yuan, Y., Liu, J., et al.; **Chen, S.**. Research Journey of Generative Protein Modeling. *NeurIPS 2024 Workshop on Advancements in Medical Foundation Models*.

- **Chen, S.**, Wang, J., Sun, L., et al. A quick paster type of soluble nanoparticle microneedle patch for the treatment of obesity. *Biomaterials*, 2024.
- Zhao, X., Zhou, Y., Kwak, W., et al.; **Chen, S.**. A reconfigurable and conformal liquid sensor for ambulatory cardiac monitoring. *Nature Communications*, 2024.
- **Chen, S.**, Manshaii, F., Fan, X., et al. Creating breathable wearable bioelectronics using three-dimensional liquid diodes. *Matter*, 2024.
- **Chen, S.**, Xu, S., Fan, X., et al. Advances in 2D materials for wearable biomonitoring. *Materials Science and Engineering: R: Reports*, 2025.
- **Chen, S.**, Fan, X., Duan, Z., Luo, Y., & Chen, J. A magnetically programmable mesoporous nanoreactor. *Nature Nanotechnology*, 2025.

Full publication list: ResearchGate

HONORS

UCLA Bioengineering Department Fellowship (2024–2025); Excellent Graduate, Peking University (2023); Scholarship for Outstanding Medical Students (2021); Award for Scientific Research, Peking University (2021); Japan Daiichi Pharmaceutical Scholarship (2018).

TECHNICAL SKILLS

Machine learning & computation: Python, PyTorch, COMSOL Multiphysics, ImageJ, AIVIA, Fusion 360, MeshLab, Arduino.

Experimental platforms: soft magnetoelastic devices, Hall-sensor arrays, microneedle fabrication, 3D printing, phantom and in vivo tumor models, materials and mechanical characterization.

TEACHING

Teaching Assistant, BE166/266 Wearable Bioelectronics; BE176 Biocompatibility.